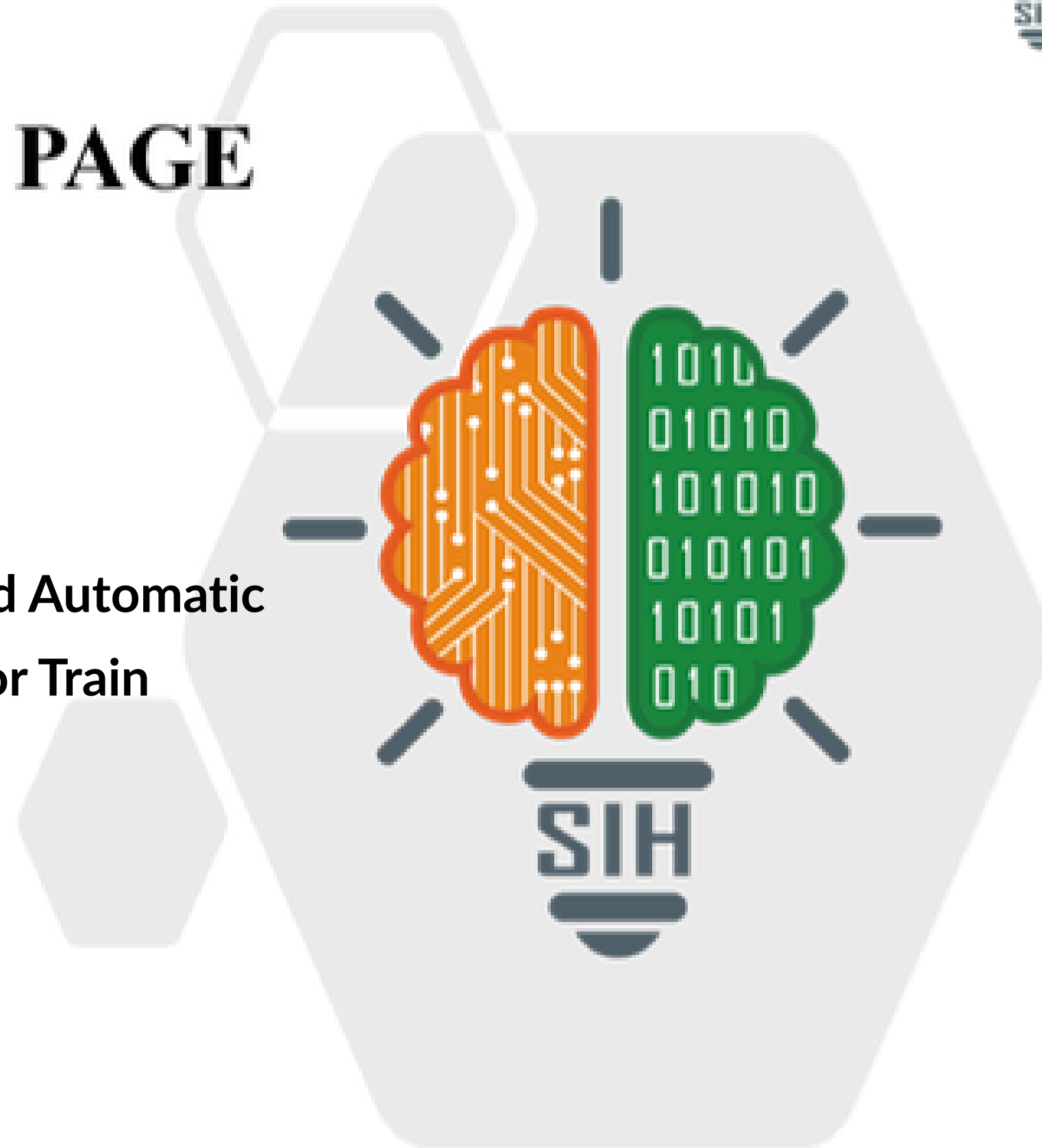
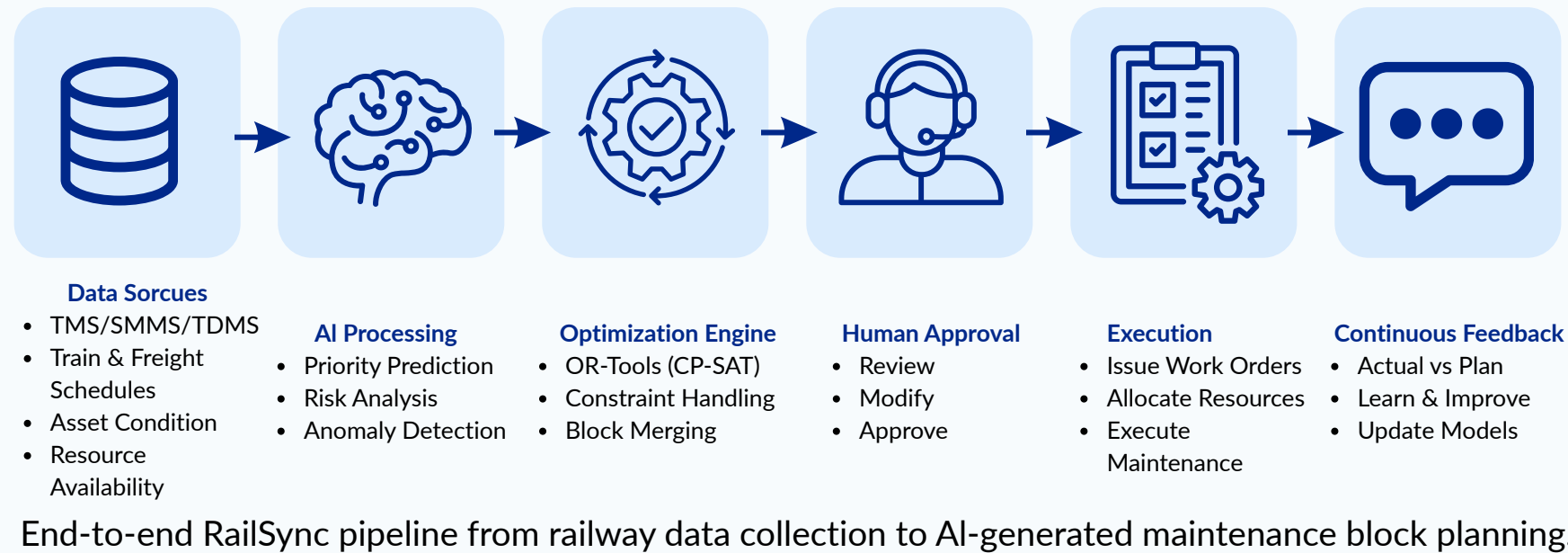


TITLE PAGE

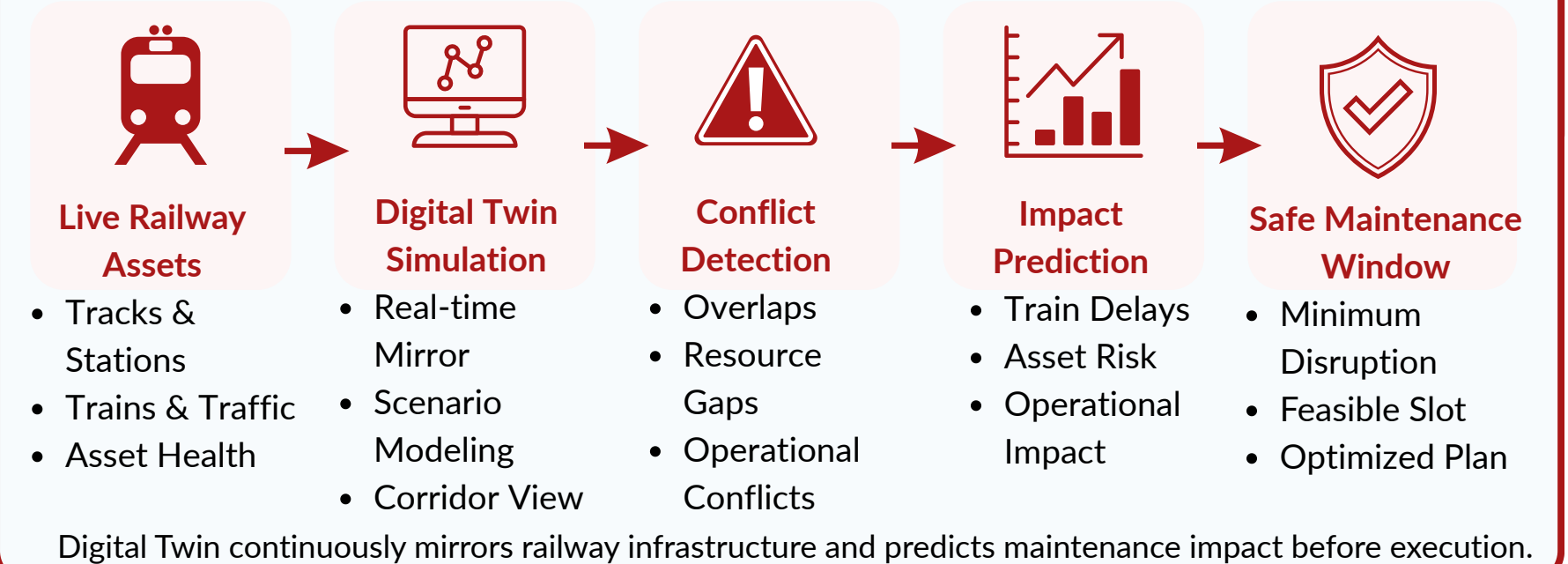
- **Problem Statement ID –** SIH26027
- **Problem Statement Title-** AI-Powered Automatic Block Planning to Maximize Asset Availability for Train Operations on Indian Railways
- **Theme-** Transportation & Logistics
- **PS Category-** Software
- **Team ID-**
- **Team Name (Registered on portal)** RailGorithm



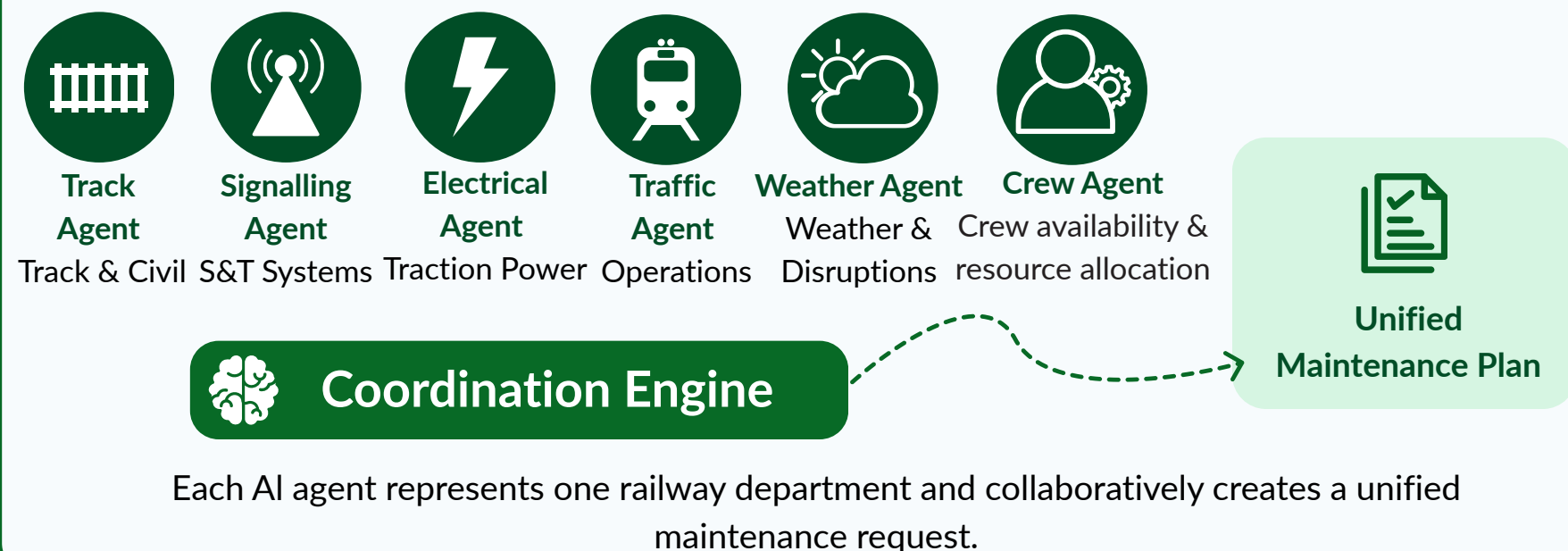
1. END-TO-END RAILS SYNC AI WORKFLOW



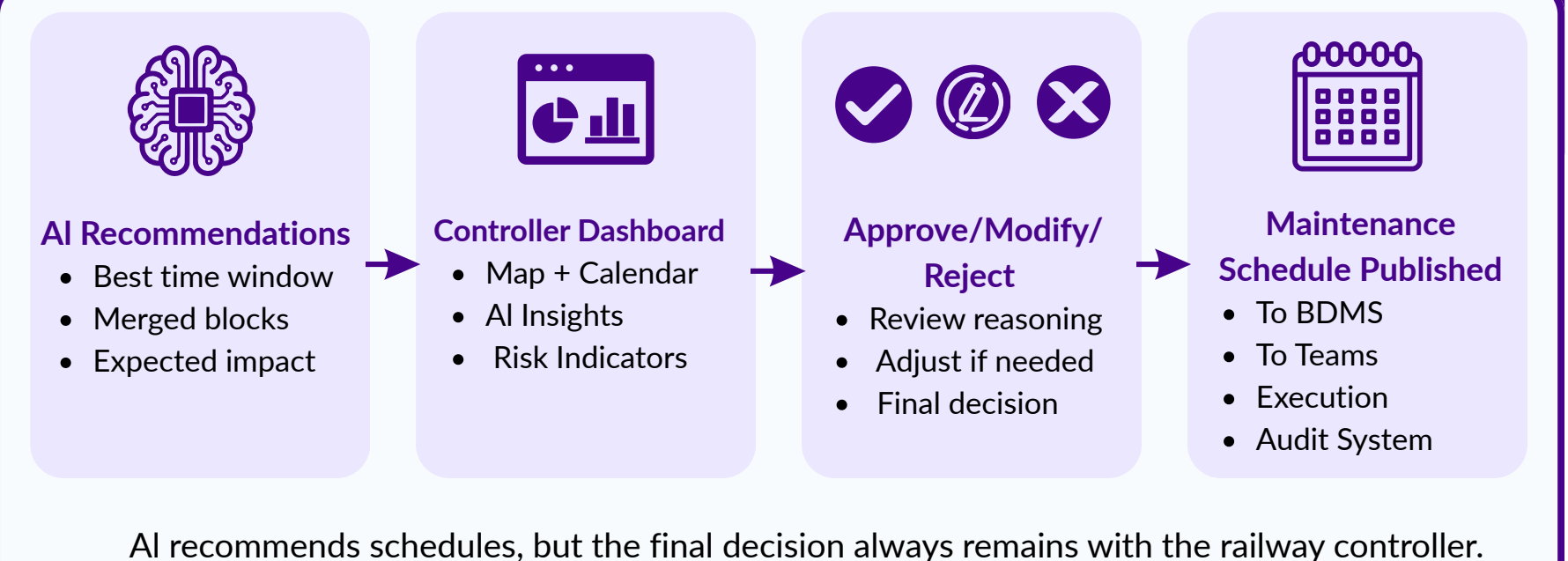
2. DIGITAL TWIN RAILWAY CORRIDOR



3. MULTI-AGENT AI COORDINATION ENGINE



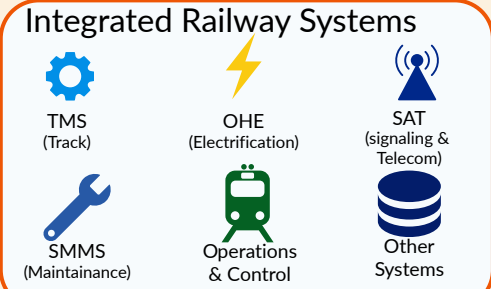
4. RECOMMENDATION ENGINE WORKFLOW





End-to-End AI Pipeline for Intelligent Maintenance Planning

1 Data Integration Layer
Unified Railway Data Hub
Converts fragmented railway databases into a single source of truth for AI planning.



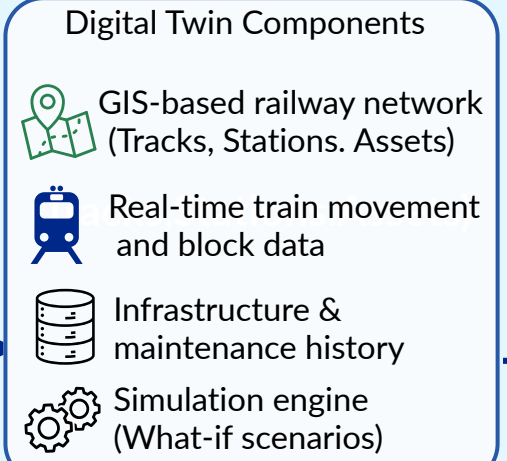
Data Processing Pipeline

- FastAPI APIs collect data
- Apache Kafka streams real-time updates
- Pandas & GeoPandas clean, normalize and merge
- PostgreSQL (records)
- TimescaleDB (time-series)

Key Benefits

- Eliminates siloed databases
- Synchronizes multi-department data
- Provides consistent inputs to AI models

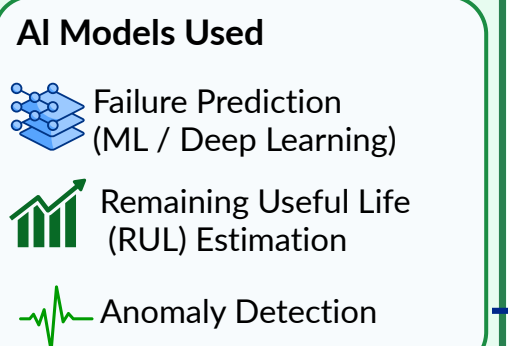
2 Digital Twin - Railway Corridor
Live Virtual Model of the Network
Creates a dynamic digital replica of the railway network before maintenance scheduling.



How It Helps

- Simulates maintenance impact before execution
- Identifies corridor conflicts in advance
- Predicts passenger delay and freight disruption
- Calculates infrastructure availability after maintenance clustering

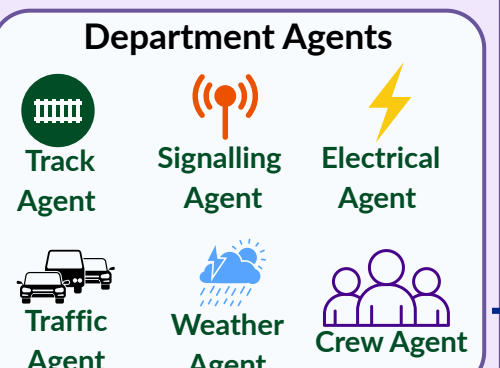
3 AI Intelligence Layer
Predict Maintenance Priorities
Uses machine learning to predict failure risks and prioritize maintenance instead of manual scheduling.



Priority Score Inputs

- Failure Probability
- Remaining Useful Life (RUL)
- Criticality index
- Traffic Density
- Overdue Maintenance
- Historical Failure Frequency

4 Multi-Agent AI Coordination
With Collaborative Agents
Uses intelligent agents to represent each railway department and enable joint decision-making.



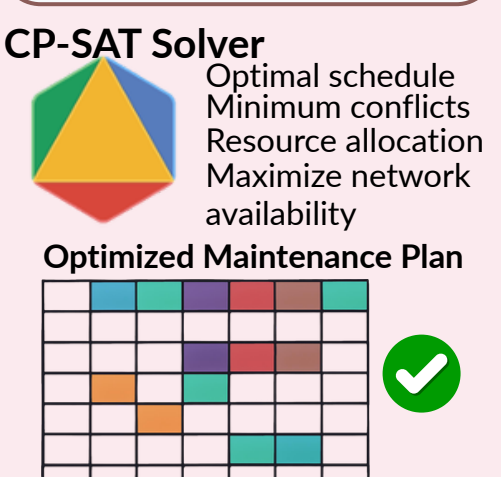
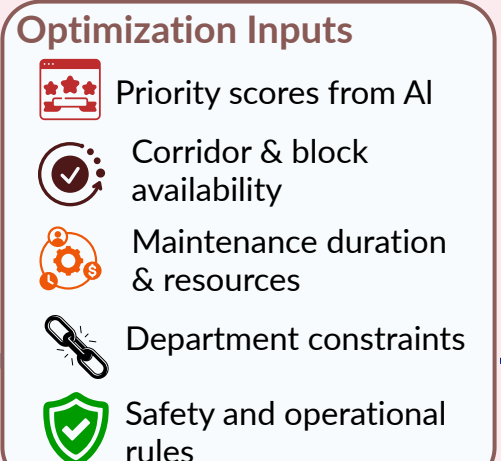
Multi-Agent Workflow

- Share maintenance plans
- Negotiate conflicts
- Find common time windows
- Suggest joint possessions

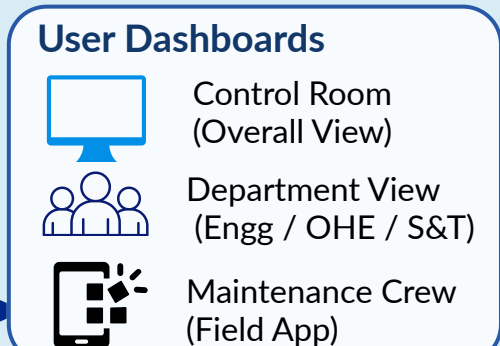
Key Benefits

- Cross-department coordination
- Shared maintenance possessions
- Reduced infrastructure downtime
- Automatic conflict negotiation

5 CP-SAT Optimization Engine
Optimal & Feasible Schedule
Uses Google OR-Tools (CP-SAT) to generate the best maintenance plan considering multiple constraints.



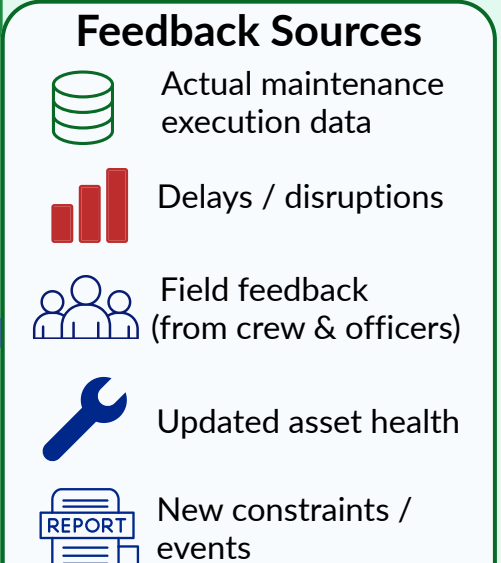
6 Dashboard & Notifications
Real-time Visibility & Alerts
Provides role-based access to optimized plans with real-time alerts and notifications



Key Features

- Interactive corridor map
- Maintenance schedules
- Real-time alerts & notifications
- What-if scenario analysis
- Reports & audit logs

7 Feedback & Continuous Learning
Improve with Real-world Data
Learns from execution outcomes to improve prediction accuracy and planning over time.



Model Improvement

- Retrain ML model
- Adjust priority scoring
- Improve optimization rules
- Better decision accuracy



FEASIBILITY ANALYSIS



Technical Feasibility

- Uses proven AI/ML, OR-Tools CP-SAT and multi-agent systems.
- Uses open-source tools and existing computing infrastructure.



Data Feasibility

- Required data available with Indian Railways (TMS, SMMS, TDMS, COA, Timetable, FOIS).
- Accessible via APIs / standard interfaces.



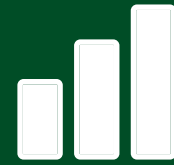
Implementation Feasibility

- Modular architecture supports phased deployment.
- Integrates with existing systems without replacing them.



Operational Feasibility

- Decision support; final control remains with railway operators.
- User-friendly UI and explainable AI support adoption.



VIABILITY



Economic Viability

- Better block utilization and coordinated maintenance reduce delays and costs.
- Optimized crew/resource allocation saves cost.



Operational Viability

- Integrated Engineering, TRD and S&T scheduling improves reliability.
- Minimizes passenger and freight disruption.



Scalability

- Pilot on one corridor/division; scale to zonal and national networks.
- Modular multi-agent framework supports new features and data.



Sustainability

- Lower fuel use, reduced idling and improved asset life.
- Supports a greener, more reliable railway network.



CHALLENGES & RISKS AND MITIGATION STRATEGIES



Data quality & completeness

- Risk: Missing/erroneous logs and inconsistent data.
- Mitigation: Validation, cleaning, augmentation and phased integration.



Real-time constraints

- Risk: Optimization must run within short operational windows.
- Mitigation: Efficient solvers, heuristics and incremental re-optimization.



Rare and extreme events

- Risk: Limited history for accidents, disasters and extreme weather.
- Mitigation: Synthetic scenarios, digital twin and continuous updates.



Human trust and adoption

- Risk: Controllers may hesitate to rely on AI recommendations.
- Mitigation: Explainable AI and human-in-the-loop decisions.

A. EXECUTIVE IMPACT STATEMENT

Railway infrastructure maintenance is essential for safety, reliability and asset performance, but maintenance blocks temporarily restrict railway operations.



Our solution — Railway Maintenance Block Optimizer — transforms this decentralized process into a data-driven, AI-assisted and optimization-based decision-support system.

It integrates Engineering, Traction and S&T requests with train schedules, asset condition and resource availability to generate optimized block plans.

SHIFT FROM

- Reactive
- Department-wise
- Manual Planning



TO

- Predictive
- Integrated
- AI-Assisted

D. IMPACT ON INDIAN RAILWAYS



- ✓ Fewer repeated shutdowns
- ✓ Better departmental coordination
- ✓ Better use of maintenance windows
- ✓ Reduced infrastructure downtime
- ✓ Faster maintenance & task visibility
- ✓ Better resource utilization
- ✓ Systematic weekly & monthly planning

B. REDUCTION IN DUPLICATE MAINTENANCE BLOCKS

Different departments often request overlapping blocks for the same corridor. Our optimizer merges them into a single coordinated block.

TYPICAL SCENARIO (MANUAL PLANNING)

	DEPARTMENT	REQUESTED BLOCK	IMPACT OF MANUAL PLANNING
	Engineering	10:00-12:00	Multiple shutdowns
	Traction Distribution	11:00-13:00	Higher disruption
	Signal & Telecommunication	12:00-14:00	Inefficient use of window

OUR SOLUTION (OPTIMIZED COORDINATED BLOCK)

Optimizer evaluates location, duration, requirement, resources, train operations and constraints to merge these into a single coordinated block.

KEY OUTCOMES

- ✓ Fewer shutdowns
- ✓ Less infrastructure downtime
- ✓ Better of maintenance windows
- ✓ Better inter-department coordination
- ✓ Higher maintenance completion
- ✓ Systematic weekly & monthly planning

E. TECHNOLOGY & DECISION SUPPORT LAYER



AI/ML

Predicts asset health, failure probability & maintenance priority



Optimization (OR-Tools)

Finds the best block combination considering constraints, deadliness & resources



Digital Twin & Simulation

Tests feasibility, visualizes impacts & validates the optimized plan



Operational Dashboard

Real-time view of asset health, blocks, availability & key operational insights.



Human-in-the-Loop

Domain experts review, adjust and approve the final block plan

G. END-TO-END IMPACT CHAIN (HOW VALUE IS CREATED)



1. Predict
Predict asset health & maintenance needs



2. Prioritize
Prioritize tasks based on risk, deadlines & operational impact



3. Coordinate
Bring all departments & operations to a platform



4. Optimize
Create the best block plan using AI & optimization



5. Simulate
Validate through Digital Twin & impact analysis



6. Human Approve
Experts review & approve the recommended plan



7. Execute
Execute safely monitor in real-time & learn continuously

C. IMPACT ON TARGET AUDIENCE



Passengers/Daily Commuters

More reliable train operations, fewer delays and cancellations, safer journeys.



Freight Operators

Smoother goods movement, fewer restrictions, improved reliability of supply chains.



Emergency Services

Faster response through better traffic flow and reduced disruptions.



Maintenance Crews

Better planning, safe access, adequate resources, less last-minute pressure.



Railway Planners & Decision Makers

Data-driven visibility, AI-assisted recommendations, transparent and auditable planning.



Government & Society

Safer infrastructure, efficient public investment, reduced congestion and environmental benefit.

F. SCALABILITY & FUTURE SCOPE



Scalable Across Indian Railways

Designed for modular deployment can be extended from one route to multiple divisions, zones and the entire network.



Future-Ready Platform

Supports integration with GIS, IoT sensors and national digital infrastructure initiatives.

ECONOMIC & OPERATIONAL IMPACT



Fewer Possessions

Compatible maintenance activities are clustered together. Instead of multiple isolated closures, jobs are completed in a single possession.



Less Train Disruption

The CP-SAT optimization engine considers active train traffic to intelligently select the least disruptive maintenance windows.



Optimized Resource Utilization

Maintenance crews and heavy machinery are deployed far more efficiently across sections, reducing transport time and idle labor.

ACADEMIC RESEARCH FOUNDATION



Lidén & Joborn (2017)

Optimization model for integrated planning of railway traffic and network maintenance.

[Click Here](#)



Zhang et al. (2019)

Microscopic optimization model integrating train timetabling and track maintenance scheduling.

[Click Here](#)



Budai et al. (2006)

Scheduling Preventive Railway Maintenance Activities & clustering tasks efficiently.

[Click Here](#)



AI & Digital Twin Research

Prasad & Januar / Fatima et al. (2026): Frameworks for predictive maintenance priority scoring.

[Click Here](#)



DATA



PREDICT



BUNDLE



OPTIMIZE



LEARN

LONG-TERM SYSTEM VALUE



Predictive Shift

Data indicates high failure risks, allowing maintenance to be planned before a disruptive failure occurs.



Decision Support

Lays the foundation for a Digital Twin, providing scalable corridor-level planning and faster re-planning.

DATA & TECHNOLOGY STACK



Indian Railways Manuals

Works & P.Way Manuals defining domain rules, safety workflows, and possession constraints.



Google OR-Tools

CP-SAT solver engine used to model timetables against maintenance windows.



DataMeet Datasets

Trains.json & Stations.json used for prototype network visualization.